NAME: ______________________ **PERIOD:** ________ **DATE:** ____________

Screen Time, Two Deletions

AP Statistics · Unit 1 · Topics 1.1–1.7 (CED 1.1–1.7) · remediation sheet

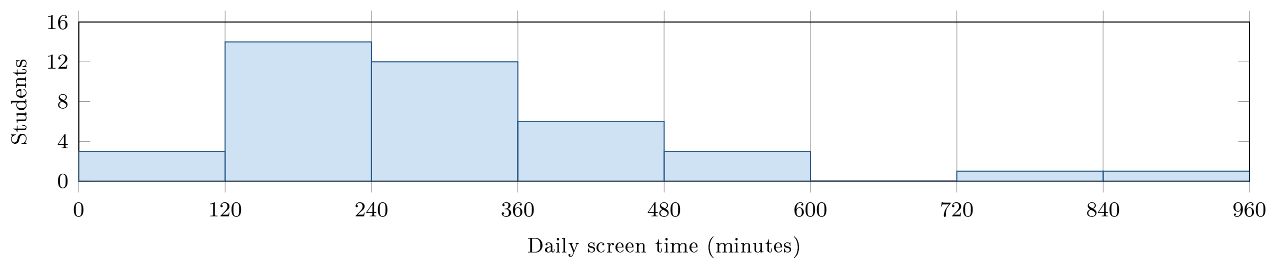
[STAR] TODAY'S PROBLEM

A school nurse recorded, for a random sample of 40 students, three things: grade level (Freshman or Senior), the device they use most (Phone, Laptop, or Console), and daily screen time in minutes from the phone's own report. The principal asked one question: *Is too much screen time a concern for our students?* The nurse's first report said only: *12 of the 40 are over 240, that is 30%, so yes.*

Group sizes: 25 freshmen, 15 seniors. Over 240 minutes: 9 freshmen, 3 seniors.

The two largest values in the histogram (a 900-minute and a 720-minute entry) came from two students who logged a whole weekend by mistake. The nurse deleted both and re-computed: the mean fell from 205 to 178 minutes; the median moved from 176 to 174 minutes; the headline became *10 of 38 are over 240, that is 26%.*

All 40 students, before any deletion



FIRST TAKE — before we discuss (5 min)

Before any calculation: does the nurse's first sentence answer the principal's question? Name one thing the sentence leaves out.

[BOOK] RULES YOU MUST USE (NO ANSWERS HERE)

A number means nothing until you say WHO was measured, WHAT was measured, the UNITS, and the QUESTION it answers.
Individuals are the things measured; variables are the characteristics recorded. A label or ID is not a measurement.
Groups of different sizes are compared with relative frequencies (percents), never raw counts alone.
A full description names shape, center, variability, AND unusual features. Some statistics resist unusual values; some do not.

Work (a)–(d) from the rules box:

DOK 1 **(a)** Fill the table: who or what are the individuals, and for each recorded variable, what ONE value looks like and whether it is categorical or quantitative. Checklist: an individual is one thing measured; a name, label, or ID is not a measurement; quantitative values could sensibly be averaged.

Individuals:	____________________	
Variable	**One value looks like**	**Categorical / Quantitative**
Grade level		
Device used most		
Daily screen time		

DOK 2 **(b)** A senior says: *Nine freshmen are over 240 and only three seniors are, so freshmen are three times as bad.* Fix the comparison. Frame: *Of the _____ freshmen, _____ are over 240, which is _____%. Of the _____ seniors, _____ are over 240, which is _____%. So the fair comparison is ____________ because the groups ____________.* Then explain in one sentence why the count 9 versus 3 can mislead here.

DOK 2 **(c)** Deleting the two values moved the mean by 27 minutes and the median by 2 minutes. Which of these two statistics is resistant to unusual values? Explain what in the histogram caused the big move, using the words *resistant* and *unusual values*. Do not say which one the nurse should report yet.

DOK 3 **(d)** ⋆ Judge the nurse's trimmed report in three to five sentences. Before you write, check: (1) WHO, WHAT, and the UNITS behind the 26%; (2) what question 26% answers, and whether that is the principal's question; (3) which center to report, tied to your answer in (c); (4) whether deleting the two mistaken values was responsible and what the report must SAY about them. Use the frame below if it helps.

Frame: I would report ________ because ____________; removing the two values changed ________ but not ________, so the principal should ____________.

EXIT — turn this in

One sentence I will add to my next written answer so a number is never left without context: ______